

EXPERIMENT 1: WATER JUG PROBLEM

Aim

To write a Python program to solve the Water Jug Problem using Artificial Intelligence techniques.

Objective

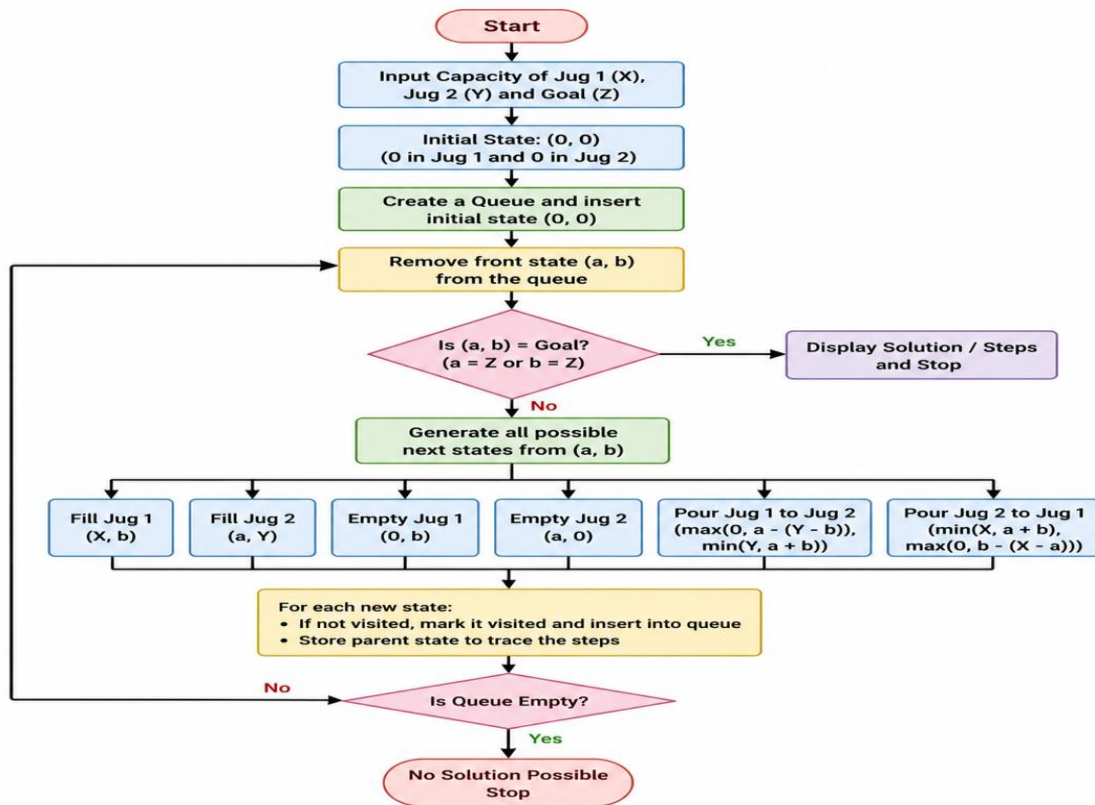
- To understand the state-space representation of AI problems.
- To implement a problem-solving technique for finding the solution to the Water Jug Problem.
- To learn how search algorithms can be used to reach a goal state.

Algorithm

1. Start.
2. Define the capacities of two water jugs.
3. Define the initial state and goal state.
4. Generate possible operations:
 - Fill a jug.
 - Empty a jug.
 - Pour water from one jug to another.
5. Check whether the generated state is valid.
6. Repeat the process until the goal state is reached.
7. Display the sequence of states.
8. Stop.

Flowchart:

EXPERIMENT 1: WATER JUG PROBLEM FLOWCHART



Code (Python)

```
from collections import deque
```

```
def water_jug(jug1, jug2, target):
```

```
    visited = set()
```

```
    queue = deque([((0, 0), [])])
```

```
    while queue:
```

```
        (a, b), path = queue.popleft()
```

```
        if (a, b) in visited:
```

```
            continue
```

```
        visited.add((a, b))
```

```
path = path + [(a, b)]
```

```
if a == target or b == target:
```

```
    return path
```

```
states = [
```

```
    (jug1, b),    # Fill Jug 1
```

```
    (a, jug2),    # Fill Jug 2
```

```
    (0, b),       # Empty Jug 1
```

```
    (a, 0),       # Empty Jug 2
```

```
    (a-min(a, jug2-b), b+min(a, jug2-b)), # Pour A to B
```

```
    (a+min(b, jug1-a), b-min(b, jug1-a)) # Pour B to A
```

```
]
```

```
for state in states:
```

```
    if state not in visited:
```

```
        queue.append((state, path))
```

```
return None
```

```
jug1 = 4
```

```
jug2 = 3
```

```
target = 2
```

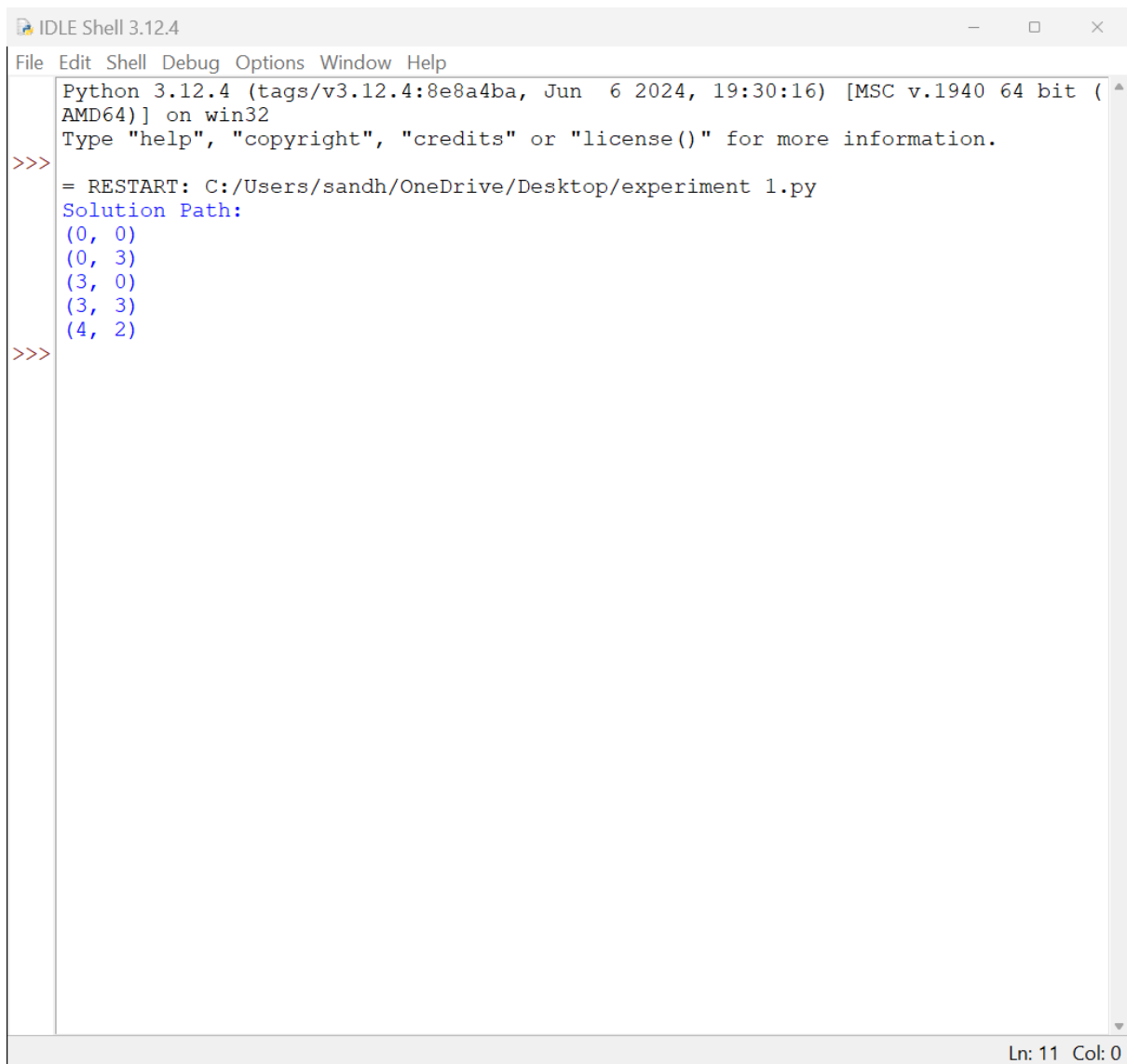
```
solution = water_jug(jug1, jug2, target)
```

```
print("Solution Path:")
```

```
for step in solution:
```

```
    print(step)
```

Output



```
IDLE Shell 3.12.4
File Edit Shell Debug Options Window Help
Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun  6 2024, 19:30:16) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/sandh/OneDrive/Desktop/experiment 1.py
Solution Path:
(0, 0)
(0, 3)
(3, 0)
(3, 3)
(4, 2)
>>>
Ln: 11 Col: 0
```

Result

Thus, the Python program to solve the Water Jug Problem was executed successfully, and the required amount of water was obtained.

Conclusion

The experiment helped in understanding AI problem-solving methods and state-space search techniques. The Water Jug Problem was solved successfully using Python by exploring possible states and finding the goal state.